Introduction to Artificial Intelligence

Konstantinos I. Bougioukas, MSc, PhD

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I propose to consider the question,
“Can machines think?”

Alan Turing in 1950

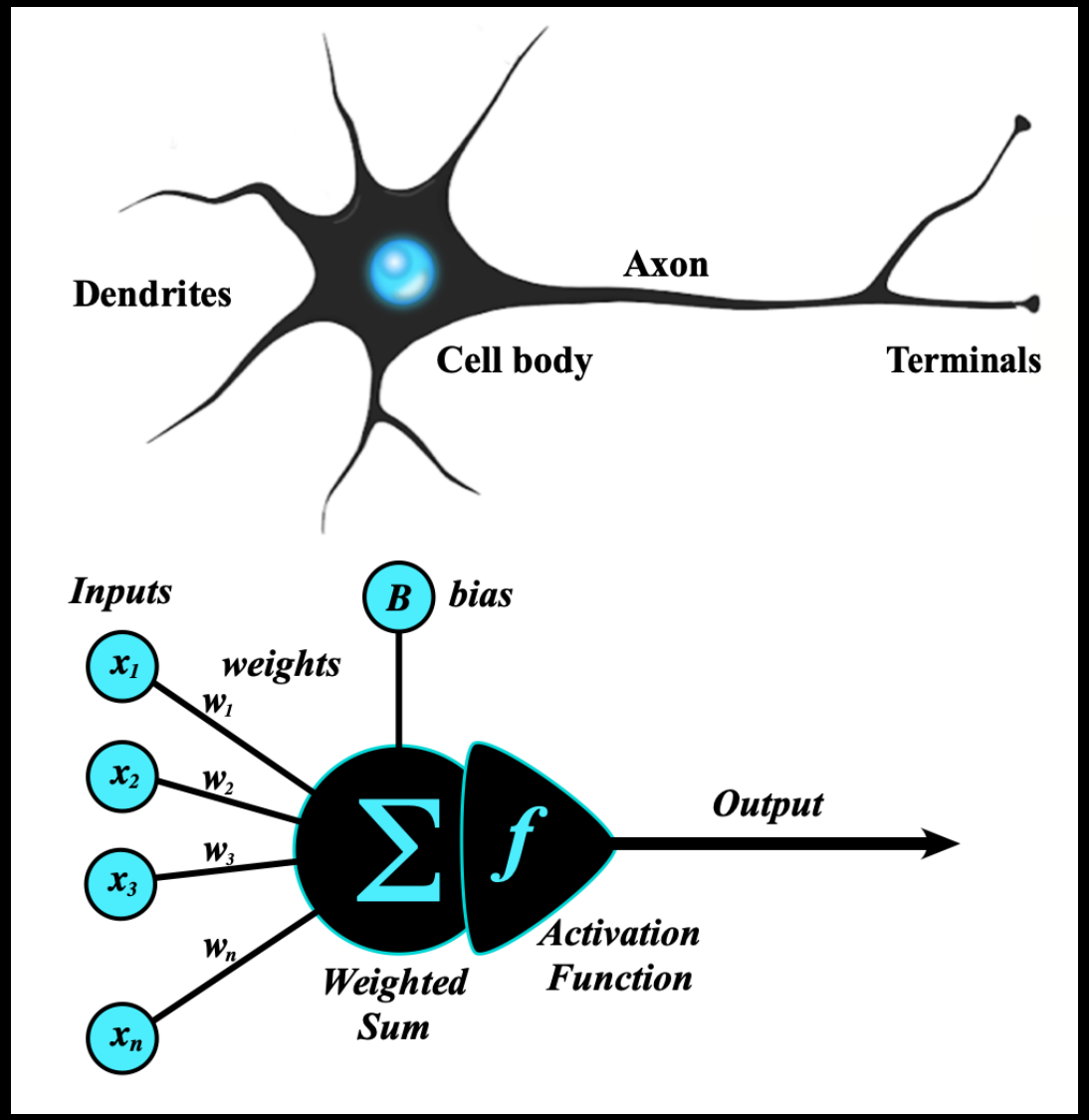
Turing, A. M. (1950). Computing machinery and intelligence. *Mind*, 59(236), 433–460.

Human Brain

- The average adult human brain consists of **billion** of neurons
- **Each neuron** can have on average 1000-10.000 synapses
- Information processing is mainly in the **cerebral cortex**, the outer layer of the brain (grey matter)
- Cognitive functions, including language, abstract reasoning, and learning and memory, represent the most complex brain operations to define in terms of neural mechanisms

The brain is a dynamic information processing system that evolves its structure and functionality in time through information processing at different hierarchical levels: quantum, molecular (genetic), single neuron, ensemble of neurons, cognitive, and evolutionary.

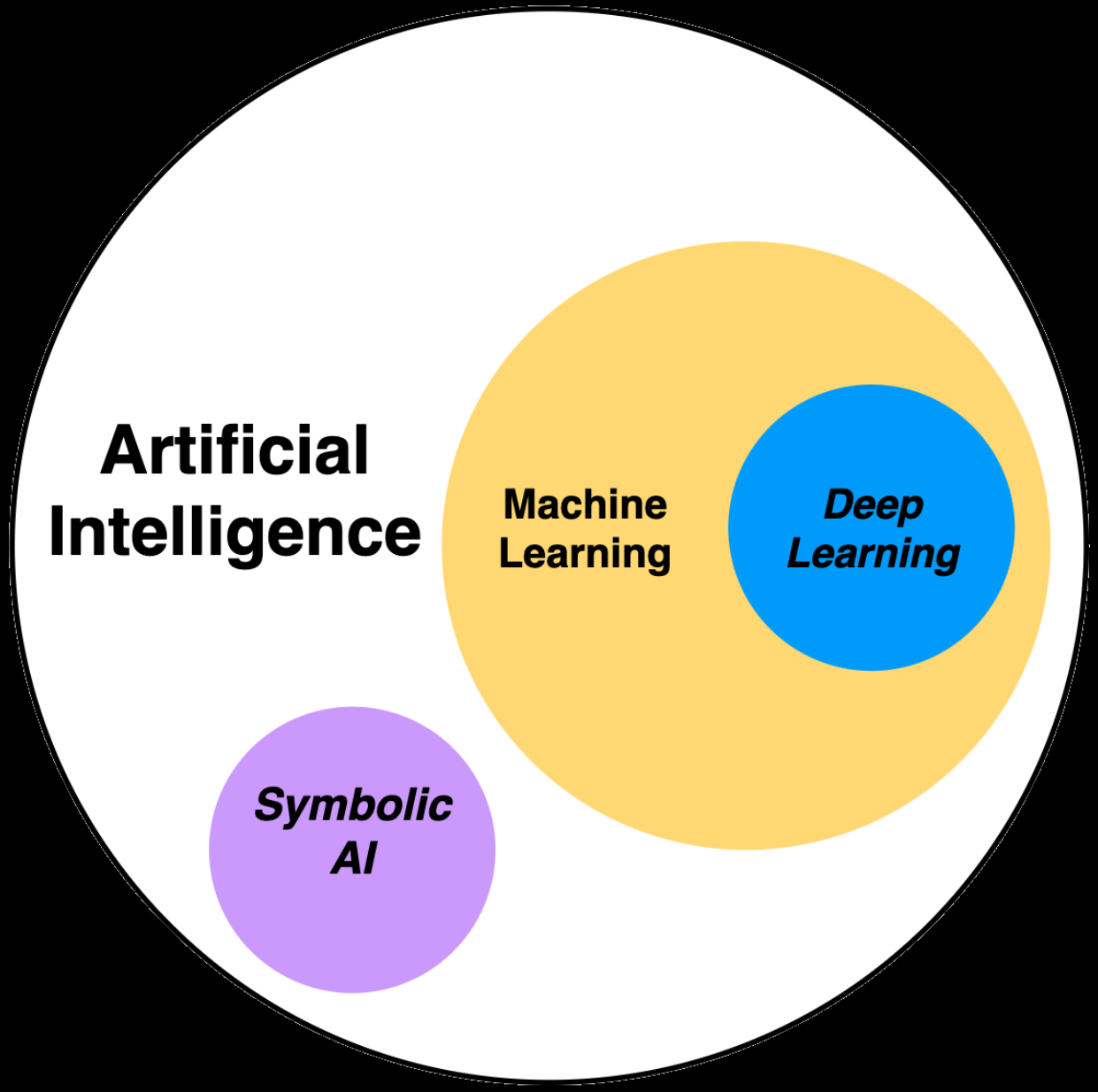
Biological Vs Artificial neurons



A biological neuron

An artificial neuron (perceptron)

The AI Hierarchy



Artificial Intelligence (AI)

AI is focused on creating machines or systems that can perform tasks typically requiring human-like intelligence, such as reasoning, learning, decision-making, language processing, and problem-solving.

Symbolic AI

Symbolic AI represents knowledge using explicit symbols and formal rules, enabling the application of logical operations to these symbols.

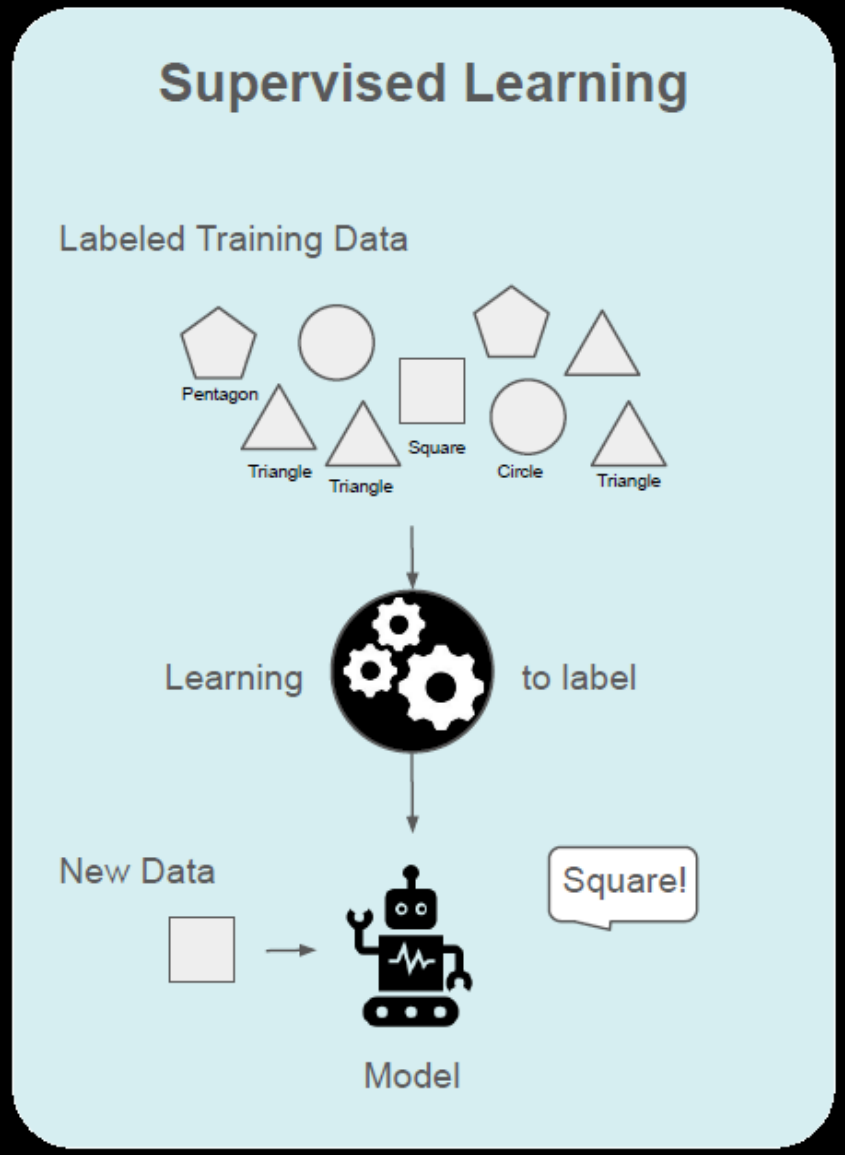
Machine Learning (ML)

ML learns to perform tasks from data without being explicitly programmed.

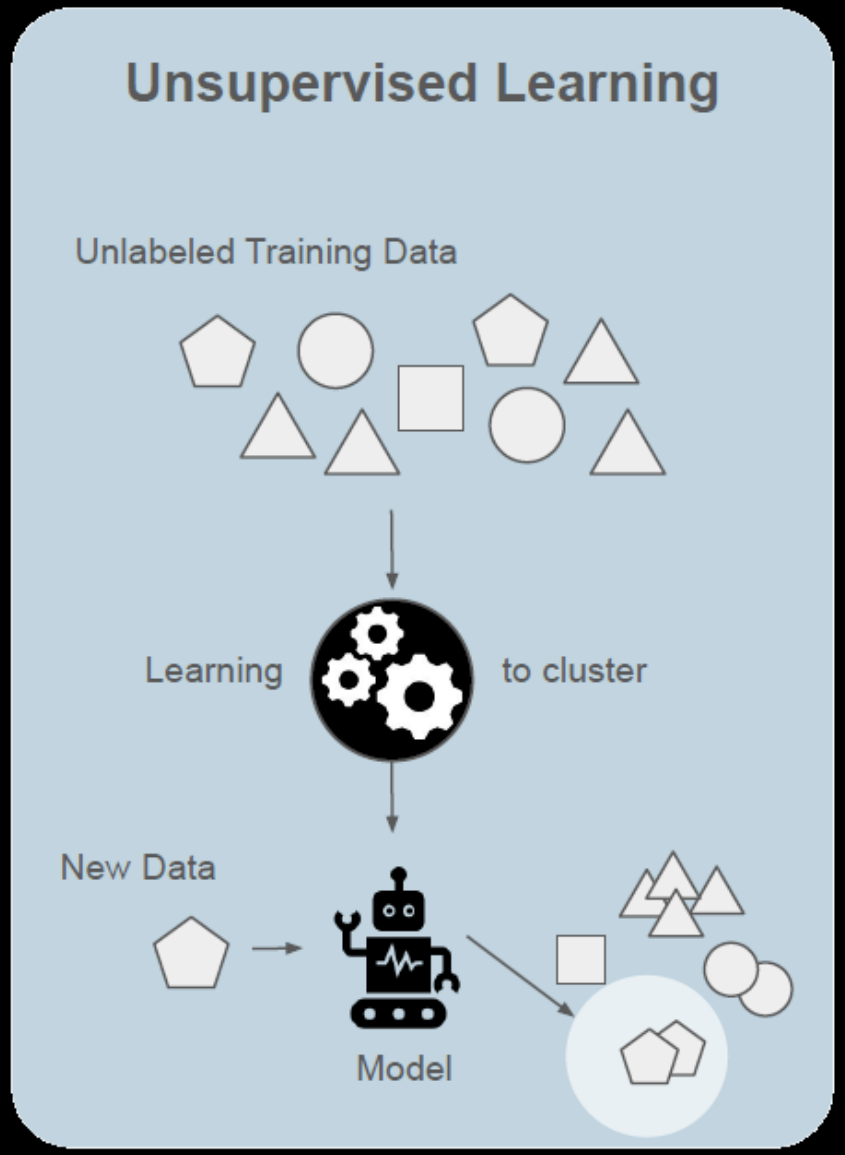
Deep Learning (DL)

DL extracts patterns from data using deep (multi-layered) neural networks.

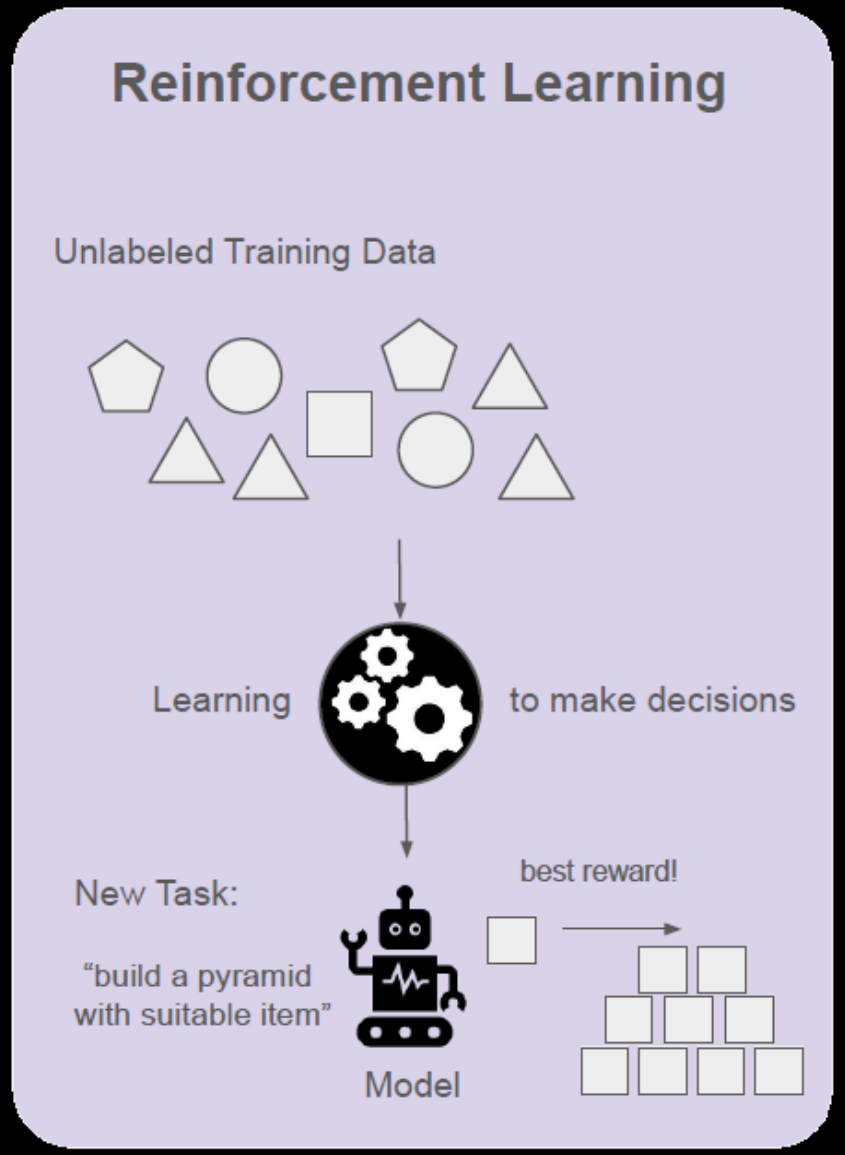
Learning methods of ML



Learns from labeled data

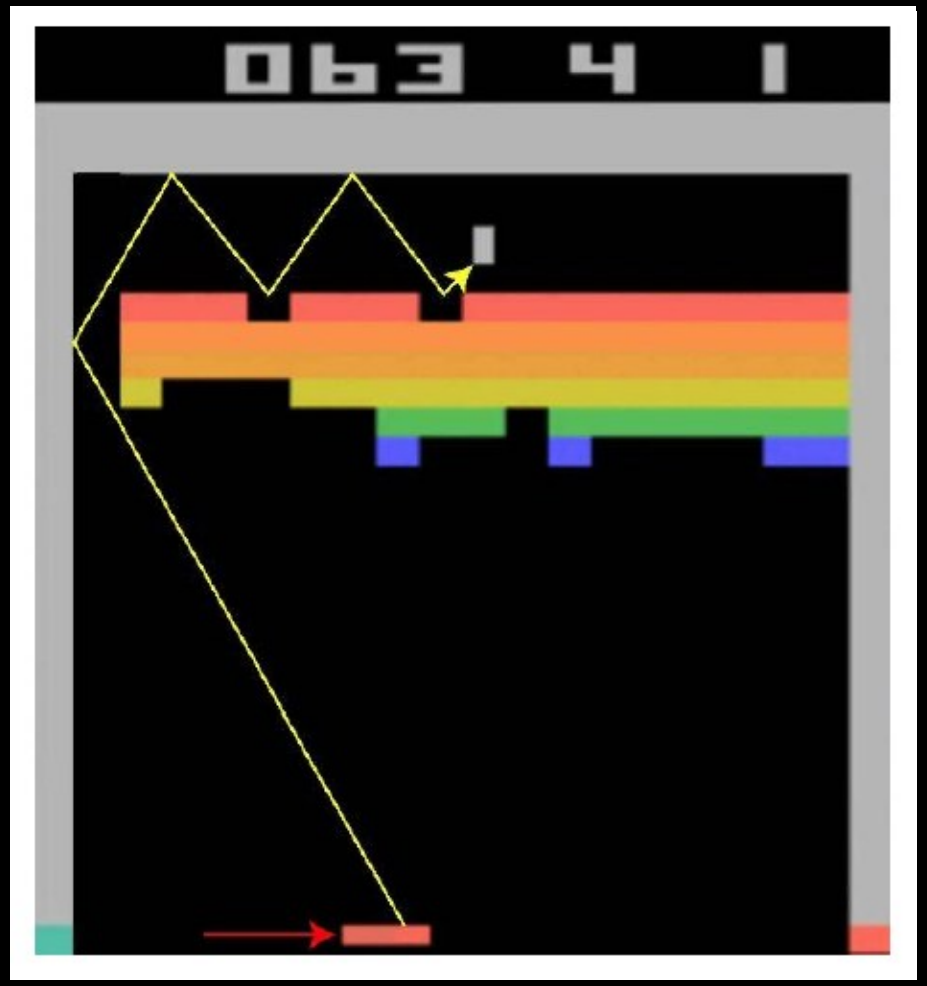


Involves unlabeled data, finds hidden patterns



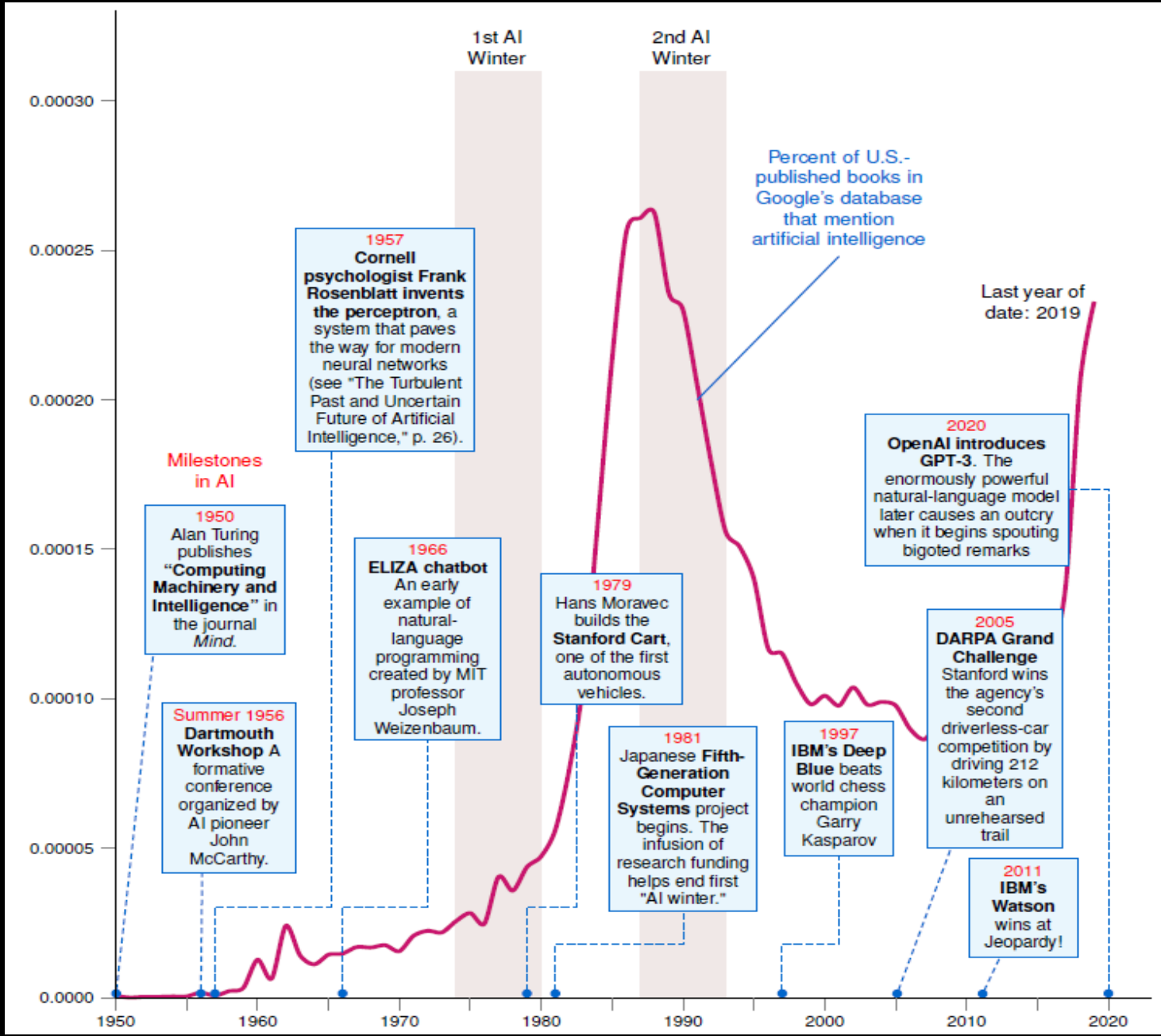
Learns by trial and error, rewards and penalties

Atari game Breakout (2014)



After learning to play the Atari game Breakout from scratch, using deep reinforcement learning to maximize the score, the DeepMind AI discovered the optimal strategy:

Drilling a hole through the leftmost part of the brick wall and letting the ball keep bouncing around behind it, amassing points very rapidly.



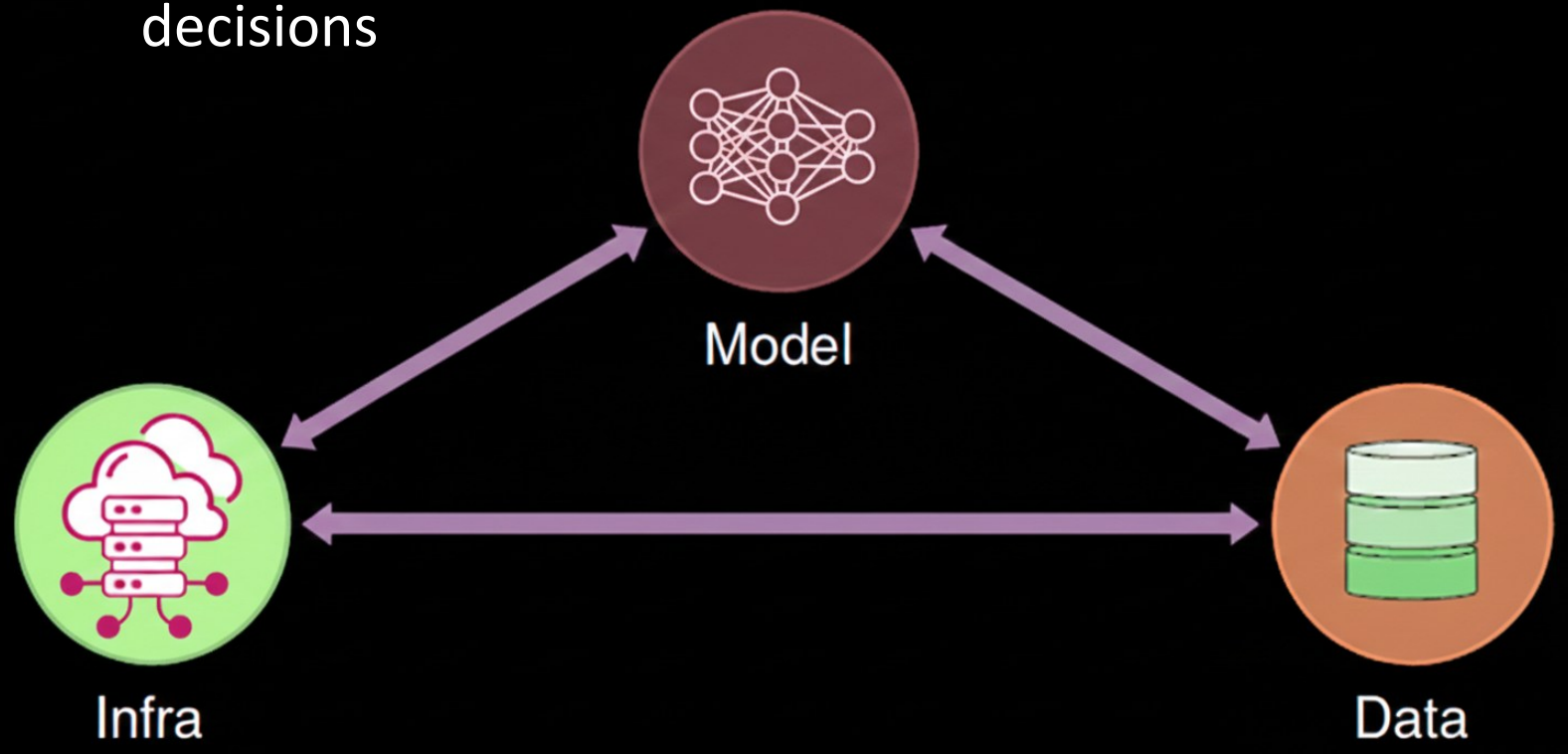
Historical Evolution of AI Paradigms

- Early AI research focused on symbolic reasoning and rule-based systems (Symbolic AI)
- Modern AI leverages data-driven approaches like neural networks to achieve increasingly complex tasks

This progression exposes a shift from hand-coded intelligence to learned intelligence, marked by milestones such as the perceptron, deep blue, and large language models like GPT-3

Components of AI Systems

Mathematical models, algorithms, and methods that learn patterns from data to make predictions or decisions



Hardware and software infrastructure that enables training, serving, and operation of models at scale

Processes and infrastructure for collecting, storing, processing, managing, and serving data for both training and inference

Classification of Artificial Intelligence

Type	Alternate Names	Scope/Capabilities	Current Status	Examples/Notes
Artificial Narrow Intelligence (**ANI**)	Weak AI, Narrow AI	Single task, high proficiency	Widely implemented	Siri, Alexa, Deep Blue, Google Search, movie recommendations
Artificial General Intelligence (**AGI**)	Strong AI, General AI	Human-level intelligence, multi-domain	Theoretical ?	Sophia robot
Artificial Super Intelligence (**ASI**)	Superintelligence	Surpasses human intelligence, including creativity, social wisdom, and scientific innovation	Hypothetical	Science fiction, ethical debates